eGov Agent

TECHNICAL WHITEPAPER

A Feasible Agentic Orchestration Layer for eGovPH

Using the government connections that already exist - and completing the work for the citizen.

THESIS eGovPH already provides the digital front door and shared rails. eGov Agent adds a consent-aware execution layer that can understand a request, coordinate approved services, and return one verifiable outcome without moving agency ownership of records.

VERSION 1.0 / 22 July 2026

Prototype build 31ea953 / Technical evaluation edition

01 / EXECUTIVE SUMMARY

Executive summary

eGov Agent is a proposed orchestration layer for eGovPH: a citizen asks for an outcome in plain language, the system identifies the responsible services, asks for narrowly scoped consent, executes through approved government APIs, and returns a synchronized status, document, appointment, report acknowledgement, or receipt.

The proposition is deliberately incremental. It reuses existing platforms - including eGovDX, National ID eVerify, eGov Pay, eReceipt, eReport, and agency or LGU systems of record - rather than building a parallel identity, payment, or records platform. DICT already presents these components as part of the national e-government ecosystem.[1]

What changes for the citizen

- One request replaces manual discovery across agency portals.
- Verified profile facts and previously approved documents can be reused with provenance instead of re-entered.
- Cross-agency work is represented as one auditable workflow, not a chain of opaque redirects.
- Payments are completed through eGov Pay and are shown as final only after agency-side verification.
- High-risk actions remain consented, policy-checked, reversible where possible, and attributable.

Recommendation

PROCEED WITH A BOUNDED PILOT Start with one read-only service, one appointment or filing service, one eGovPay transaction, and one LGU eReport workflow. Require sandbox APIs, a Privacy Impact Assessment, contract tests, explicit consent receipts, and measurable service-level objectives before adding more agencies.

Evidence standard

This paper separates implemented user experience from simulated integrations and proposed production components. The current repository is a strong interaction prototype; it is not a connected government production system. No live agency credentials, resident database, payment settlement, or emergency dispatch is claimed.

02 / PROTOTYPE BOUNDARY

What exists today - and what does not

The audited build is a Next.js 16.2.10 and React 19.2.4 client application. Its interaction model is implemented; its government outcomes are deterministic simulations. This is appropriate for a hackathon proof of experience, provided the production gap is explicit.

Capability	Current build	Production requirement
Conversational service discovery	**Implemented**	Replace scripted intent matching with policy-governed routing and evaluated models.
Agent activity trace	**Implemented**	Populate from real distributed traces and adapter events.
File stamps, previews, uploads	**Implemented**	Object storage, malware scanning, encryption, retention, DLP.
Personal memory	**Simulated**	Provenance-first store, consent, expiry, correction, deletion, source reconciliation.
National ID authentication	**Simulated**	PSA relying-party onboarding and NIDAS/eVerify integration.
Agency records and appointments	**Simulated**	Agency-approved APIs, delegated authorization, sandbox and UAT.
eGovPay and eReceipt	**Simulated**	Server-side order creation, signed callback/polling, ledger reconciliation.
eReport dispatch	**Simulated**	Authoritative routing directory, agency queues, acknowledgements, escalation rules.
Backend, database, queue	**Not present**	Government-hosted control plane and durable workflow engine.

Auditable implementation footprint

Source	Purpose
app/agent/brain.ts	Typed cards, scripted intent-to-plan mapping, trace steps, vault references.
app/agent/page.tsx	Conversation UI, attachments, voice entry, agent execution animation.
app/agent/forms.ts	Printable forms, appointment passes, simulated receipts and eReport acknowledgement.
app/agent/memory/page.tsx	Source-labelled personal context interface.
app/agent/vault/page.tsx	Document vault interactions, file previews and uploads.

CRITICAL BOUNDARY The current UI must never be connected directly to agency APIs. All tokens, signatures, policy checks, payment verification, document issuance, and audit events belong in a server-side government control plane.

03 / REFERENCE ARCHITECTURE

A thin orchestration layer over existing rails

The target design keeps agency systems authoritative. eGov Agent stores only workflow state, user-approved context, and cryptographic or audit evidence required to complete and explain a transaction. It does not replicate agency registries into a general-purpose AI database.

Production reference architecture

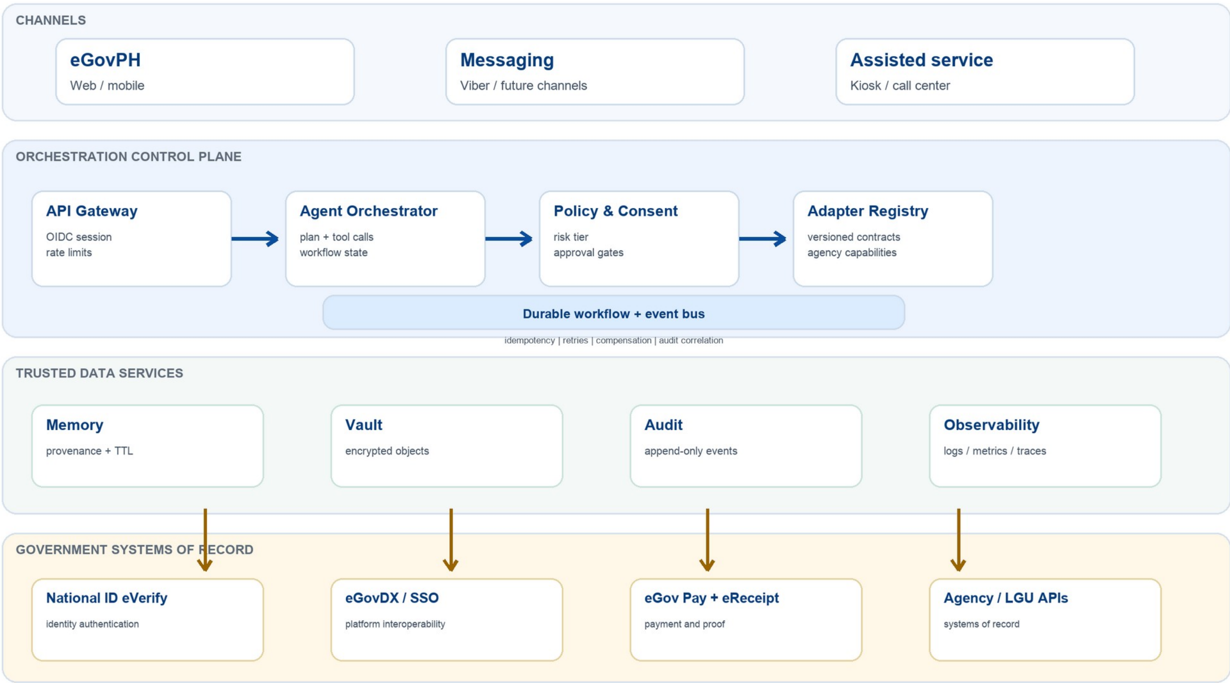

Figure 1. Production reference architecture. Arrows represent authenticated, policy-governed API calls; agency systems remain the systems of record.

Core design decisions

- Control plane, not monolith: the orchestrator coordinates identity, policy, adapters, and workflow state; it does not absorb agency business logic.
- Adapter isolation: every agency integration is versioned, rate-limited, observable, and removable without changing the conversation layer.
- Durable execution: long-running work uses a workflow engine and event bus so retries do not duplicate bookings, payments, or reports.
- Minimal AI privilege: models propose structured plans; a deterministic policy engine decides which tools and data scopes are allowed.

04 / TRANSACTION LIFECYCLE

From request to verifiable outcome

Government transactions are distributed workflows. Network timeouts, partial success, manual review, and delayed agency processing are normal conditions. The design therefore treats a conversation as a durable state machine rather than one model response.

High-risk transaction lifecycle

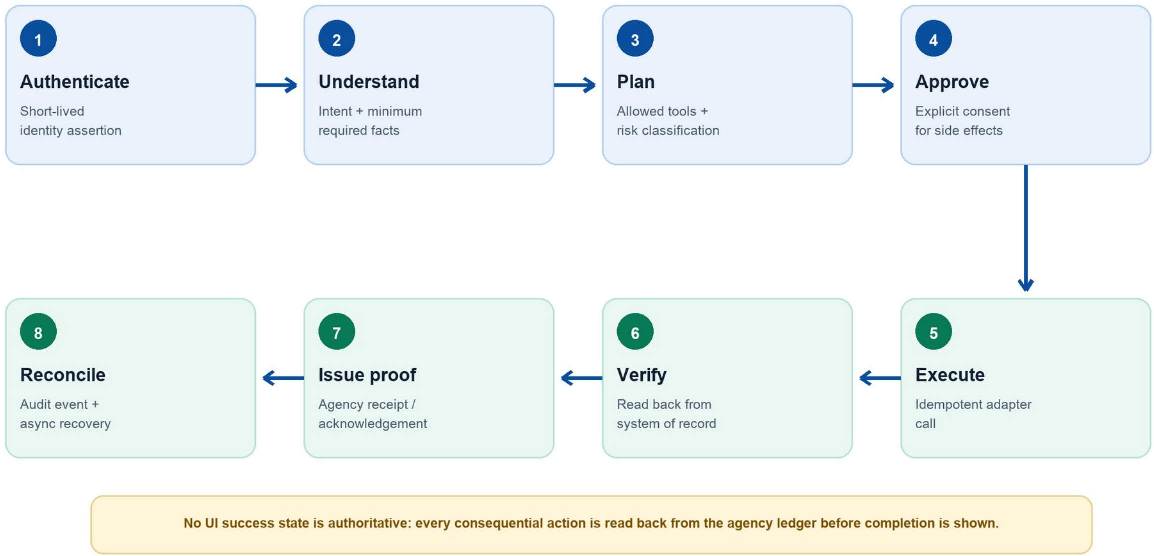

Figure 2. Every consequential action passes through authentication, policy, explicit approval, idempotent execution, authoritative read-back, and reconciliation.

Execution invariants

Invariant	Engineering rule
No duplicate side effects	Client supplies an idempotency key; adapter persists result before acknowledging.
No model-issued authority	Only allow-listed tools can cause effects; policy evaluation is deterministic.
No inferred payment success	Success requires a verified gateway event plus agency ledger confirmation.
No silent scope expansion	A new agency, purpose, data category, or fee requires renewed approval.
No lost partial state	Workflow checkpoints and compensating actions survive process restarts.

DELIVERY SEMANTICS Distributed systems cannot promise exactly-once delivery across independent agencies. The feasible contract is at-least-once messaging with idempotent agency operations, deterministic deduplication, and reconciliation.

05 / INTEGRATION CONTRACTS

A stable adapter contract for uneven agency systems

Agencies differ in authentication, data models, turnaround times, and legacy constraints. eGov Agent normalizes the orchestration contract while preserving the agency-specific implementation behind each adapter.

```
POST /v1/actions/execute
Idempotency-Key: 01J...
Authorization: Bearer <delegated-token>

{
  "action": "appointment.reserve",
  "agency": "DFA",
  "subject_ref": "opaque-user-ref",
  "consent_receipt": "cr_...",
  "input": {"site_id": "...", "slot_id": "..."}
}

202 Accepted
{"operation_id": "op_...", "status": "PENDING", "poll_after_s": 5}
```

Concern	Contract
Authentication	OIDC/OAuth 2.1 where available; short-lived delegated token; mTLS between services.
Idempotency	Required for every side effect; scope key to subject + agency + action + request hash.
Async operations	202 + operation ID; webhook or polling; terminal states are explicit.
Errors	Stable codes: validation, consent, auth, rate limit, unavailable, manual review, rejected.
Versioning	Semantic adapter versions; deprecation window; contract tests in agency sandbox.
Evidence	Correlation ID, policy decision ID, consent receipt, agency reference, timestamps.

Real onboarding path

DICT's published Citizen's Charter describes eGovPH integration as a government-to-government process involving a Letter of Intent, eGovPH Request Form, Memorandum of Understanding, technical integration, internal testing, UAT, and deployment; timing depends on client readiness.[8] The implementation plan in this paper follows that path rather than assuming public, self-service APIs.

FEASIBILITY PRINCIPLE Pilot only with agencies that can provide an accountable product owner, sandbox or test double, canonical service rules, and a named incident/escalation path.

06 / IDENTITY, MEMORY, AND VAULT

Personalization without a surveillance graph

National ID eVerify is a realistic identity rail, but integration requires relying-party onboarding. PSA describes eVerify as an authentication service for government and private relying parties and notes regulatory and technical onboarding, including API access.[3][4] eGov Agent should consume a scoped identity assertion, not store biometrics or the PhilSys registry.

Recommended memory architecture

Store	Use	Why
Relational profile facts	Address city, language, notification preference	Queryable, correctable, source-linked, easy to expire.
Limited relationship graph	Household, authorized representative, document-to-case links	Useful where relationships matter; avoid a universal knowledge graph.
Vector index	Semantic retrieval over user-approved documents	Find relevant text; never the authoritative fact store.
Agency references	Opaque IDs and last-known state	Point back to systems of record rather than copying registries.

Every remembered fact should carry: subject, value, source, purpose, confidence, captured_at, expires_at, consent scope, sensitivity, and supersedes. Facts synchronized from agencies are not editable as truth; the citizen can flag them for correction at the source.

Vault controls

- Encrypted object storage with envelope encryption and per-tenant keys managed by a government KMS.
- Malware scanning, MIME validation, content-disarm policy, and quarantine before any model or adapter sees a file.
- Signed, short-lived download URLs; watermarking for generated previews; no public object paths.
- Retention by purpose and document type, with legal hold and user deletion workflows where allowed.
- Document provenance: issuer, hash, verification status, valid-from/to, and each transaction that reused it.

ANSWER TO THE KNOWLEDGE-GRAPH QUESTION Use a graph selectively for real relationships. A graph should not become the primary store for all personal context; a hybrid model is easier to govern, correct, delete, and reconcile with authoritative agencies.

07 / PAYMENTS AND ERECEIPT

eGovPay integration that can survive reconciliation

The public eGovPay site describes both a Payment Gateway API and Payment Links, with cards, online bank transfers, and e-wallets.[2] The production design should use a hosted or government-controlled checkout so the agent never handles payment credentials.

Authoritative payment flow

1. The agency creates an order server-side with service code, assessed amount, expiry, citizen reference, and idempotency key.
2. eGovPay returns a hosted checkout reference. The citizen reviews the exact agency, service, amount, and refund rules before authorization.
3. A signed webhook and/or server-to-server status query reports gateway settlement. Redirect success is never sufficient.
4. The adapter validates merchant, order, amount, currency, state transition, timestamp, and replay nonce.
5. The agency posts the payment to its ledger and returns the official receipt or eReceipt reference.
6. Only then does eGov Agent mark the transaction paid and display the immutable receipt copy.

State	Meaning	Citizen-facing behavior
CREATED	Order exists; no payment	Show fee and expiry.
PENDING	Authorization or settlement in progress	Do not resubmit; allow status refresh.
PAID_UNPOSTED	Gateway paid; agency ledger pending	Show processing; start reconciliation.
PAID	Agency ledger confirmed	Release official proof and continue service.
FAILED / EXPIRED	No settlement	Allow a new order with a new idempotency key.
REFUND_PENDING / REFUNDED	Reversal workflow	Show agency-owned refund reference.

NON-NEGOTIABLE The agent may format and present a receipt, but it must not invent an official receipt number. The issuing agency or the designated eReceipt service must return that identifier after verified posting.

08 / EREPORT

A credible eReport workflow: assist, confirm, dispatch, track

The prototype's flooding scenario demonstrates the intended experience: accept a photo and description, infer likely incident type, resolve location, identify responsible desks, ask the citizen to review, dispatch once, and synchronize acknowledgements. Production safety requires the AI to assist triage - never to impersonate a dispatcher or claim acknowledgement without an agency event.

Stage	Automation	Required guardrail
Intake	Extract description, time, media metadata	Disclose metadata use; strip unnecessary EXIF on onward sharing.
Location	Geocode and jurisdiction match	Citizen confirms pin; retain confidence and resolver version.
Triage	Suggest category and severity	Human/agency rules override model; no definitive emergency diagnosis.
Routing	Match primary and secondary responders	Authoritative service directory; deduplicate by time/place/evidence.
Dispatch	Create agency work items	Idempotent fan-out; per-agency acceptance or rejection captured.
Tracking	Aggregate status and ETA	Only display agency-supplied status; timestamp every update.
Closure	Ask for outcome feedback	Agency closes official case; citizen may dispute or reopen.

Report state model

```
DRAFT -> CONFIRMED -> DISPATCHING -> DISPATCHED_PARTIAL|DISPATCHED
      -> ACKNOWLEDGED -> IN_PROGRESS -> RESOLVED -> CLOSED
Any state -> REJECTED|DUPLICATE|CANCELLED; every transition records actor, reason, and timestamp.
```

Public-safety safeguards

- Surface the official emergency channel when immediate danger is indicated; the form must not delay emergency contact.
- Rate-limit and abuse-detect without silencing legitimate repeated reports during disasters.
- Blur faces and plates in previews by default; provide unredacted evidence only to authorized responders when lawful and necessary.
- Design for degraded connectivity: resumable uploads, SMS acknowledgement, offline draft, and eventual dispatch.

PROTOTYPE TRUTH The Mandaluyong CDRRMO, barangay, MMDA, and DPWH statuses shown in the current experience are simulated. Production must render each acknowledgement from an authenticated agency callback.

09 / SECURITY AND PRIVACY

Security model for a high-trust government agent

The Data Privacy Act requires reasonable organizational, physical, and technical safeguards.[5] NPC Circular 2023-06 is the current cross-sector security circular and adds privacy-by-design/default and business-continuity expectations; it superseded the older government-only Circular 16-01.[6] A PIA, data-sharing agreements, records of processing, incident response, and agency DPO involvement are preconditions, not post-launch paperwork.

Trust boundaries and data minimization

Figure 3. Raw identity, documents, credentials, and audit evidence remain inside the government control plane. Optional external models receive only approved minimum context.

Threat	Primary controls
Account takeover	Phishing-resistant MFA, device/session risk, step-up authentication for high-risk actions.
Prompt injection in files	Treat content as untrusted, isolate parsing, tool allow-list, schema validation, no instruction inheritance.
Excessive agency access	ABAC by purpose, data category, role, consent receipt, and transaction state.
Credential leakage	Server-only secrets, KMS/HSM, rotation, mTLS, workload identity, no secrets in prompts or logs.
Duplicate or replayed effects	Idempotency keys, signed nonces, monotonic state machine, reconciliation.
Insider misuse	Dual control for privileged actions, immutable audit, anomaly detection, access reviews.
Model data leakage	Minimization, redaction, approved tenancy, zero-retention terms, output DLP.

10 / RELIABILITY AND OPERATIONS

Designed for partial failure, not perfect dependencies

The control plane should remain useful when one agency is unavailable. It must preserve the citizen's request, show which step is delayed, avoid duplicate effects, and resume from a durable checkpoint when the dependency recovers.

Layer	Pattern	Pilot target
API edge	Stateless replicas, rate limits, WAF, request correlation	99.9% monthly availability
Workflow	Durable state, queue, retries with jitter, dead-letter queue	No lost accepted jobs
Adapters	Timeouts, circuit breakers, bulkheads, per-agency quotas	Failure isolated by adapter
Payments	Outbox/inbox, ledger read-back, daily reconciliation	Zero unreconciled duplicates
Data	Multi-AZ database, point-in-time recovery, encrypted backups	RPO <= 15 min; RTO <= 2 h
Observability	OpenTelemetry traces, redacted logs, SLO alerts	>= 95% transactions trace-complete

Performance budget

- Conversation acknowledgement: p95 under 800 ms before long-running work begins.
- Policy evaluation and adapter selection: p95 under 250 ms inside the control plane.
- Agency actions: reported independently with their own SLA; never hidden behind a typing animation.
- Streaming status: server-sent events or push notifications backed by persisted workflow events.

Capacity model

```
Peak requests/s = monthly sessions x turns/session x peak factor / active seconds
Queue concurrency = arrival rate x p95 dependency latency / target utilization
Storage = workflow events + audit evidence + retained documents; scale each independently.
```

OPERATIONAL OWNERSHIP Each adapter needs a runbook, dashboard, on-call owner, status page, data owner, recovery procedure, and contact at the source agency. Integration count is not a success metric if operational ownership is missing.

11 / IMPLEMENTATION AND SCALABILITY

A gated path from prototype to public pilot

Phase	Duration	Deliverables	Exit gate
0 - Mobilize	0-6 weeks	Agency owners; PIA; data inventory; service blueprint; integration request; threat model	Approved scope, legal basis, sandbox plan
1 - Read-only	6-12 weeks	eGovPH SSO/eVerify assertion; one record lookup; audit; support console	>=95% correct routing; no critical findings
2 - Transact	12-20 weeks	Consent receipts; appointment/filing; eGovPay; reconciliation; notifications	Zero duplicate effects; payment match >=99.95%
3 - eReport	16-24 weeks	LGU routing; media handling; agency queue; acknowledgement; escalation	Responder UAT; safety drill passed
4 - Scale	6-12 months	Adapter SDK; onboarding playbook; DR; multilingual and assisted channels	SLOs met for 90 days; independent review

Recommended production stack

Concern	Technology-neutral choice
Web and API	Next.js front end; typed backend service; OpenAPI 3.1; government API gateway.
Workflow	Durable workflow engine plus queue/event bus; transactional outbox.
State	PostgreSQL-compatible relational store; object storage; Redis-compatible ephemeral cache.
Identity	OIDC/OAuth 2.1, eGovPH SSO, eVerify relying-party integration, workload identity.
Security	KMS/HSM, secrets manager, WAF, SIEM, SAST/DAST/SCA, container signing and policy.
AI	Model gateway, approved local model for common tasks, optional redacted external model, evaluation harness.

AI deployment strategy

Use deterministic rules and templates for common government intents; retrieval for service requirements; compact local models for classification and extraction; larger models only for complex language tasks. Every model output is schema-validated, scored, and subject to policy. This reduces cost and blast radius while retaining a natural interface.

ACCESSIBILITY AND INCLUSION Target WCAG 2.2 AA, Filipino and English, screen-reader semantics, large-text support, low-bandwidth modes, human handoff, assisted-service channels, and printable outcomes from the first pilot.

12 / COST AND VALUE

A transparent pilot estimate - not a procurement quote

The following range is an engineering planning estimate for a 9-12 month pilot with two national agencies, one LGU, up to 50,000 monthly active users, one payment flow, and one eReport flow. It excludes agency legacy-system remediation, statutory fees, and nationwide contact-center operations.

Cost area	Planning range (PHP)	Assumption
Product and integration team	10.5M-18.0M	8-10 FTE equivalent; product, backend, frontend, QA, DevSecOps, design.
Security, privacy, legal, assurance	2.0M-4.0M	PIA, threat model, pen test, DSA/MOU support, independent review.
Cloud, observability, messaging	1.5M-3.5M	Pilot traffic, multi-AZ services, logs/traces, SMS/email; excludes subsidies.
Agency onboarding and change	1.5M-3.0M	Service mapping, sandbox/UAT, training, support runbooks.
Contingency	2.5M-5.0M	Legacy variance, compliance changes, recovery work.
Total pilot envelope	18.0M-33.5M	Range should be replaced by discovery and government procurement estimates.

Illustrative value model

Assumption	Illustrative input	Derived monthly value
Citizen time avoided	15,000 completed transactions x 45 min	11,250 citizen-hours
Agency handling reduced	15,000 x 8 min	2,000 staff-hours
Repeat data entry avoided	30,000 forms x 5 min	2,500 citizen-hours
Unit run cost	Annual operating cost / completed transactions	Track by service and channel

These are hypotheses to test, not impact claims. The pilot must measure baseline journey time, completion, office visits avoided, staff touch time, payment reconciliation effort, and accessibility outcomes before declaring benefit.

Cost controls

- Route common intents deterministically; reserve expensive models for the small complex tail.
- Cache public service metadata, never sensitive personal records.
- Charge showback by agency adapter, model tier, notification channel, and completed transaction.
- Stop integrations that cannot meet operational ownership or citizen-value thresholds.

13 / GOVERNANCE AND RISK

Risks that can make a polished agent unsafe

Risk	Why it matters	Mitigation / owner
Hallucinated eligibility or status	A confident error can deny service or create false assurance.	Rules sourced from agency; read-back; citations; agency product owner.
Over-broad consent	Citizens may approve without understanding cross-agency scope.	Purpose-specific receipts, preview, expiry, revocation; DPO.
Legacy API instability	Retries can duplicate actions or create inconsistent states.	Adapter idempotency, circuit breakers, reconciliation; integration owner.
Biased triage	Reports or assistance may be deprioritized unfairly.	Human rules, subgroup evaluation, appeal, no sole automated adverse decision.
Digital exclusion	People without modern devices or literacy can be left behind.	Assisted channels, low-bandwidth design, printable/SMS results; service owner.
Scope creep	A super-agent can accumulate excessive data and authority.	Risk tiers, architecture review, per-service approval, kill switch; steering group.
Vendor lock-in	Models or workflow services may become hard to replace.	Open contracts, model gateway, portable data, exit plan; enterprise architecture.

Decision rights

Decision	Accountable authority
Whether a service may be automated	Source agency plus DICT governance and legal/privacy review
What personal data may be processed	Personal Information Controller, DPO, approved PIA/data-sharing terms
Whether a transaction is complete	Source agency system of record
Whether a payment is settled	eGovPay evidence plus agency ledger
Whether an eReport is acknowledged/resolved	Receiving agency or LGU
Which model may be used	Government AI/security governance under approved risk tier

HUMAN ACCOUNTABILITY The agent can recommend, prepare, and execute approved steps. It cannot become the legal decision-maker for adverse eligibility, enforcement, emergency prioritization, or issuance unless a specific law and agency policy authorize that automation.

How technical judges - and production owners - should test it

Pre-production verification

- Contract tests against every adapter sandbox, including version drift and malformed responses.
- Replay tests for timeouts after success, duplicated webhooks, stale tokens, partial fan-out, and restart recovery.
- Security testing: threat model, SAST, dependency scanning, API fuzzing, penetration test, secrets audit, prompt-injection red team.
- Privacy testing: data-flow inventory, minimization, consent withdrawal, deletion/retention, subject-access response, log redaction.
- AI evaluation: intent accuracy, extraction accuracy, tool-selection precision, unsafe-action rate, bilingual cases, refusal/handoff quality.
- Accessibility and resilience: keyboard/screen reader, 200% zoom, low bandwidth, device loss, offline draft, assisted channel.

Pilot scorecard

Metric	Target / guardrail
Correct service routing	>=95% on adjudicated pilot set
End-to-end completion	>=70% for eligible, supported journeys; measure reasons for fallback
Duplicate side effects	0 confirmed duplicate bookings, reports, or charges
Payment reconciliation	>=99.95% automatically matched; all exceptions resolved within one business day
Consent comprehension	>=90% can identify agency, action, data, and fee in usability test
Critical security/privacy incidents	0; tested notification and containment procedure
Accessibility	WCAG 2.2 AA audit with no critical blockers
Citizen time	Measured reduction versus baseline, segmented by channel and accessibility need

GO-LIVE RULE A visually impressive conversation is not a readiness signal. Go live only after system-of-record verification, failure recovery, privacy/security controls, agency UAT, support ownership, and measurable citizen benefit all pass.

Conclusion

eGov Agent is technically feasible because it does not require government to start over. The necessary rails - a one-stop eGovPH channel, National ID authentication, data-exchange infrastructure, government payment capabilities, eReceipt, eReport, and agency systems - already exist in varying degrees. The missing product layer is safe orchestration: durable workflows, explicit consent, stable adapters, policy enforcement, authoritative read-back, and one citizen-facing trail.

The current prototype successfully communicates that future. Its next milestone is not a more powerful chatbot; it is a narrow, governed pilot with real agency owners, sandbox contracts, measurable outcomes, and production controls. If that pilot proves reliable, the adapter model can scale service by service without centralizing agency records or granting a model uncontrolled authority.

BOTTOM LINE Keep the conversation simple for citizens and the controls rigorous underneath. The agent should make government feel unified while preserving where identity, money, records, and legal decisions actually belong.

References

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[5] National Privacy Commission, Republic Act No. 10173 - Data Privacy Act of 2012. https://privacy.gov.ph/data-privacy-act/

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[7] Lawphil, Republic Act No. 11032 - Ease of Doing Business and Efficient Government Service Delivery Act of 2018. https://lawphil.net/statutes/repacts/ra2018/ra_11032_2018.html

[8] DICT Citizen's Charter 2025, Integration in eGovPH Application, pp. 42-45. https://cms-cdn.e.gov.ph/DICT/pdf/DICT-Citizens-Charter-2025-1st-Edition.pdf

Document basis

Repository audit: eGov Agent prototype build 31ea953, reviewed 22 July 2026. Public sources above were accessed 22 July 2026. Architecture, cost ranges, SLOs, and pilot targets are proposed engineering specifications or illustrative planning assumptions, not claims about current government production deployments.